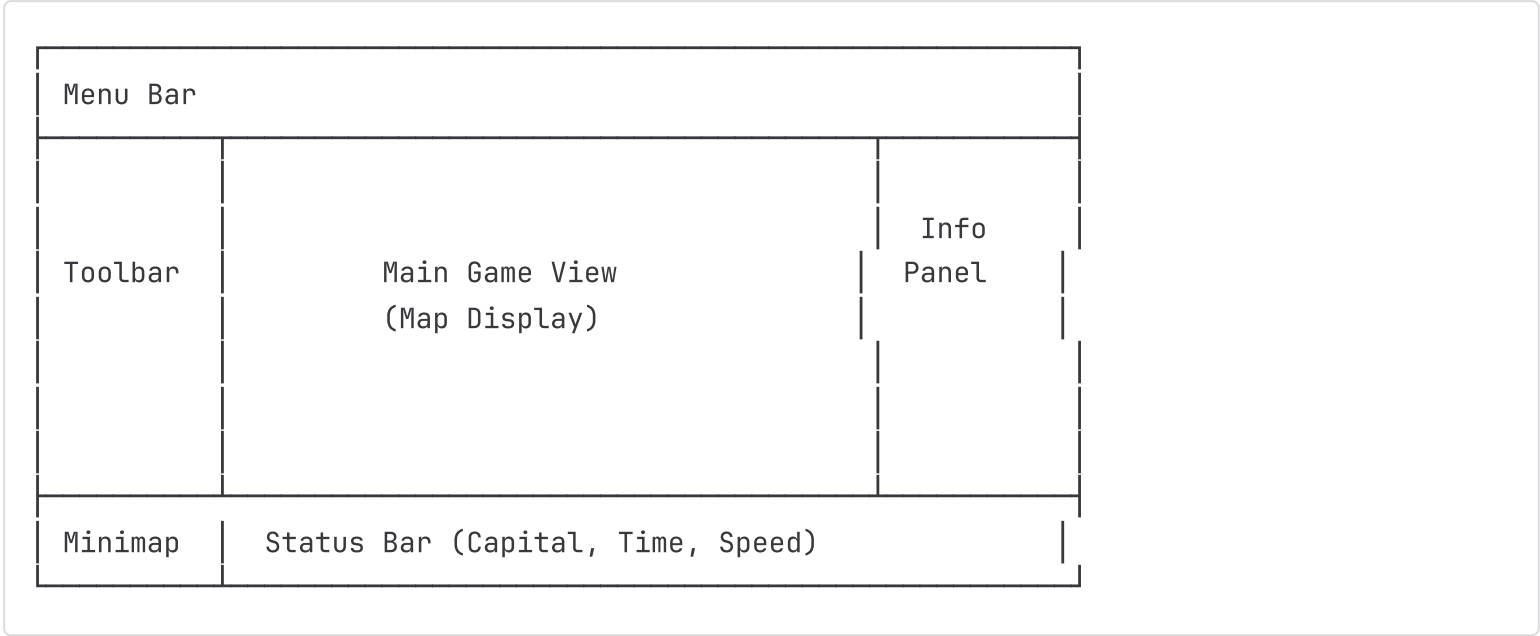


UI Plan

This page contains the user interface design plan for Mini Transport Tycoon, including wireframes, layout specifications, and interaction guidelines.

Overall Layout

Main Window Structure



Screen Layout Dimensions

- **Total Window:** 1280×720 (minimum), scalable to 1920×1080
- **Main Game View:** ~70% of width, ~85% of height
- **Toolbar:** ~15% of width, ~85% of height
- **Info Panel:** ~15% of width, ~85% of height
- **Minimap:** ~20% of width, ~10% of height
- **Status Bar:** Full width, ~5% of height

Menu Bar

File Menu

- **New Game** - Start a new game
- **Load Game** - Load a saved game
- **Save Game** - Save current game
- **Save Game As...** - Save with custom name
- **Exit** - Quit application

Game Menu

- **Pause/Resume** - Toggle game pause
- **Speed: Normal** - Set speed to normal
- **Speed: Fast** - Set speed to 2x
- **Speed: Very Fast** - Set speed to 4x
- **Settings** - Open settings dialog

View Menu

- **Zoom In** - Increase zoom level
- **Zoom Out** - Decrease zoom level
- **Reset Zoom** - Reset to default zoom
- **Toggle Minimap** - Show/hide minimap
- **Toggle Toolbar** - Show/hide toolbar

Help Menu

- **How to Play** - Open help dialog
- **Controls** - Show keyboard shortcuts
- **About** - Application information

Toolbar

Tool Selection

- **Select Tool** (default) - Select and view objects

- **Build Road Tool** - Build roads on map
- **Remove Road Tool** - Remove roads
- **Place Stop Tool** - Place stops
- **Remove Stop Tool** - Remove stops
- **Build Garage Tool** - Build garages
- **Install Traffic Light Tool** - Install traffic lights
- **Terraform Tool** - Change terrain height

Tool Display

- Icons for each tool
- Selected tool highlighted
- Tooltips on hover
- Keyboard shortcuts displayed

Main Game View

Map Display

- **Rendering:** Tile-based 2D top-down view
- **Scrollable:** Mouse drag or arrow keys
- **Zoomable:** Mouse wheel or zoom controls
- **Tile Highlighting:** Highlight on mouse hover
- **Selection:** Click to select objects

Visual Elements

- **Cities:** Colored rectangular areas with buildings
- **Facilities:** Distinct icons/shapes per facility type
- **Roads:** Lines connecting tiles, different styles for directions
- **Vehicles:** Animated sprites showing type and direction
- **Stops:** Small icons at road intersections
- **Garages:** Distinct building icon
- **Traffic Lights:** Visible at intersections with color indication
- **Terrain:** Height differences shown with shading/slopes

Interaction

- **Left Click:** Select object, place building/road
- **Right Click:** Context menu, cancel action
- **Mouse Drag:** Scroll map, draw roads
- **Mouse Wheel:** Zoom in/out
- **Hover:** Show tooltip with object information

Info Panel

Selected Object Information

When an object is selected, display relevant information:

City Selected:

- City name
- Current demands (passengers, goods)
- Connected stops
- Population (if tracked)

Facility Selected:

- Facility type and name
- Production rates
- Consumption rates
- Current inventory

Vehicle Selected:

- Vehicle type
- Current cargo/passengers
- Assigned route
- Speed and status
- Age and maintenance status

- Location

Route Selected:

- Route name
- List of stops
- Assigned vehicles
- Total distance
- Average delivery time

Stop Selected:

- Stop type
- Current cargo/passengers
- Connected routes
- Location

Garage Selected:

- Garage location
- Vehicles in garage
- Maintenance queue
- Purchase options

Traffic Light Selected:

- Location
- Current states per direction
- Configuration button

Action Buttons

Context-sensitive action buttons based on selection:

- **Purchase Vehicle** (when garage selected)
- **Create Route** (when stops selected)
- **Edit Route** (when route selected)
- **Sell Vehicle** (when vehicle selected)
- **Configure Traffic Light** (when traffic light selected)

Minimap

Display

- **Size:** ~20% of window width, ~10% of height
- **Location:** Bottom-left corner
- **Content:** Entire map in miniature
- **Viewport Indicator:** Rectangle showing current view
- **Key Elements:** Cities, facilities, major roads

Interaction

- **Click:** Jump to clicked location
- **Drag:** Pan main view
- **Hover:** Show coordinates

Status Bar

Information Display

- **Capital:** Current player capital (large, prominent)
- **Game Time:** Current game time/date
- **Game Speed:** Current speed indicator (Pause/Normal/Fast/Very Fast)
- **FPS:** Current frames per second (optional, debug mode)

Speed Controls

- **Pause Button:** Toggle pause
- **Speed Buttons:** Normal, Fast, Very Fast
- **Visual Indicator:** Current speed highlighted

Dialog Windows

Route Creation Dialog

- **Name Input:** Route name field
- **Stop Selection:** List of available stops
- **Route Preview:** Visual preview of route
- **Buttons:** Create, Cancel

Vehicle Purchase Dialog

- **Vehicle Type Selection:** List with icons and stats
- **Vehicle Information:** Speed, capacity, cost, maintenance
- **Purchase Button:** Buy selected vehicle
- **Cancel Button:** Close dialog

Traffic Light Configuration Dialog

- **Direction List:** All directions at intersection
- **Green Duration Sliders:** Per direction
- **Preview:** Visual preview of timing
- **Apply Button:** Save configuration
- **Cancel Button:** Discard changes

Settings Dialog

- **Graphics Settings:** Resolution, quality
- **Audio Settings:** Volume controls (if audio added)
- **Controls:** Key bindings
- **Game Settings:** Default speed, auto-save

Save/Load Dialog

- **File List:** Available save files
- **File Information:** Date, time, capital, vehicles
- **Load Button:** Load selected file
- **Save Button:** Save with name
- **Delete Button:** Delete selected file

Game Over Dialog

- **Message:** "Game Over - Bankruptcy" or "Victory"
- **Final Statistics:** Total income, expenses, time played
- **Buttons:** New Game, Exit

Context Menus

Map Context Menu (Right-click on empty tile)

- Build Road
- Place Stop (if road present)
- Terraform Terrain
- View Terrain Info

Road Context Menu

- Remove Road
- Place Stop
- Install Traffic Light (if intersection)

Stop Context Menu

- Remove Stop
- Create Route
- View Stop Info

Vehicle Context Menu

- View Vehicle Info
- Assign to Route
- Send to Garage
- Sell Vehicle

Route Context Menu

- Edit Route
- Delete Route
- View Route Info

Visual Design Guidelines

Color Scheme

- **Background:** Neutral gray/beige for map
- **Roads:** Dark gray/black
- **Cities:** Light blue/gray
- **Facilities:** Distinct colors per type
- **Vehicles:** Bright, distinguishable colors
- **UI Elements:** Modern, clean design

Typography

- **Headers:** Sans-serif, bold, 14-16pt
- **Body Text:** Sans-serif, regular, 11-12pt
- **Labels:** Sans-serif, regular, 10pt
- **Tooltips:** Sans-serif, regular, 9pt

Icons

- **Consistent Style:** Flat design or simple 2D
- **Size:** 16×16 to 32×32 pixels
- **Toolbar Icons:** 24×24 pixels
- **Clear Distinction:** Each icon clearly different

Animations

- **Smooth Transitions:** 200-300ms for UI transitions
- **Vehicle Movement:** Continuous, smooth
- **Loading Indicators:** For save/load operations
- **Feedback:** Visual feedback for all actions

Keyboard Shortcuts

Navigation

- **Arrow Keys:** Scroll map
- **WASD:** Scroll map (alternative)
- **Mouse Wheel:** Zoom in/out
- **Space:** Pause/Resume
- **+/-:** Increase/Decrease speed

Tools

- **1:** Select Tool
- **2:** Build Road Tool
- **3:** Place Stop Tool
- **4:** Build Garage Tool
- **5:** Install Traffic Light Tool
- **6:** Terraform Tool

Actions

- **Escape:** Cancel current action, close dialogs
- **Delete:** Remove selected object
- **Ctrl+S:** Save game
- **Ctrl+O:** Load game
- **Ctrl+N:** New game

Implementation Notes

Technology

- **JavaFX:** Recommended for modern UI
- **Scene Graph:** For efficient rendering

- **CSS Styling:** For consistent appearance
- **FXML:** For UI layout (optional)

Performance

- **Dirty Rectangle Updates:** Only redraw changed areas
- **Sprite Batching:** Group similar sprites
- **Lazy Loading:** Load resources on demand
- **Viewport Culling:** Only render visible tiles

Accessibility

- **Tooltips:** All interactive elements
- **Keyboard Navigation:** Full keyboard support
- **High Contrast:** Option for color-blind users
- **Text Scaling:** Respect system font size

Comments